Honors Geometry – Chapter 4 Test PRACTICE **WORKED KEY**

Transformations • Sections 4.1 – 4.3

Show all work in the space provided. Check your work. Write each final answer on the line. Label every image with primes.

1. Write the component form of the vector from A(4, 3) to B(−1, 5). (2)

x: 4 → −1; y: 3 → 5

→ ⟨−5, 2⟩

2. Write a rule for the translation that maps P(−4, 0) to P′(−10, −7). (2)

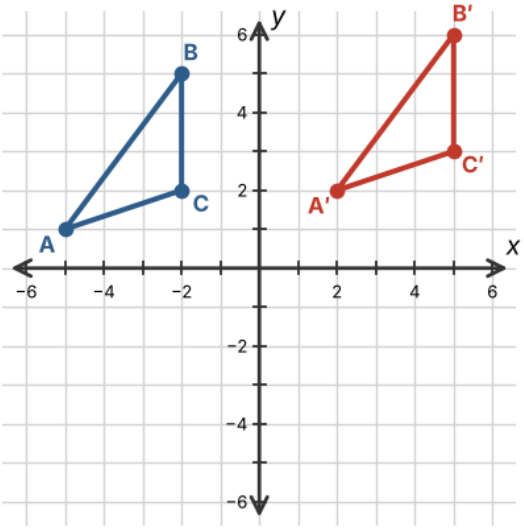
→ (x, y) → (x − 6, y − 7)

3. Q′(4, 4) is the image of Q after a translation by ⟨1, −5⟩. Find Q. (2)

undo it: ⟨−1, 5⟩

→ Q(3, 9)

4. Graph △ABC with A(−5, 1), B(−2, 5), C(−2, 2) and its image after a translation by ⟨7, 1⟩. (3)



→ A′(2, 2), B′(5, 6), C′(5, 3)

5. Translate by (x, y) → (x + 5, y − 4), then by (x, y) → (x − 2, y + 5). Write ONE rule that does both. (2)

→ (x, y) → (x + 3, y + 1)

6. Reflect the point (−7, 2) in each line. (4)

→ x-axis: (−7, −2) y-axis: (7, 2) y = x: (2, −7) y = 3: (−7, 4)

7. Reflect M(2, 3) in the line x = −1. (2)

3 units right of the line → 3 units left

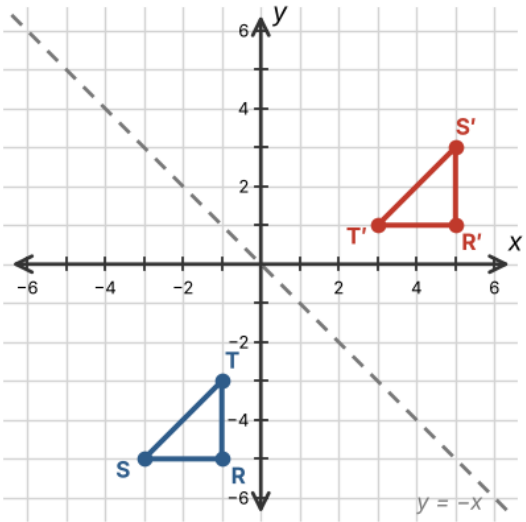
→ M′(−4, 3)

8. A reflection maps A(−6, −6) to A′(0, −6). Write the equation of the line of reflection. (2)

the line is halfway between: x = (−6 + 0) / 2

→ x = −3

9. Graph △RST with R(−1, −5), S(−3, −5), T(−1, −3) and its reflection in the line y = −x. (3)



(x, y) → (−y, −x)

→ R′(5, 1), S′(5, 3), T′(3, 1)

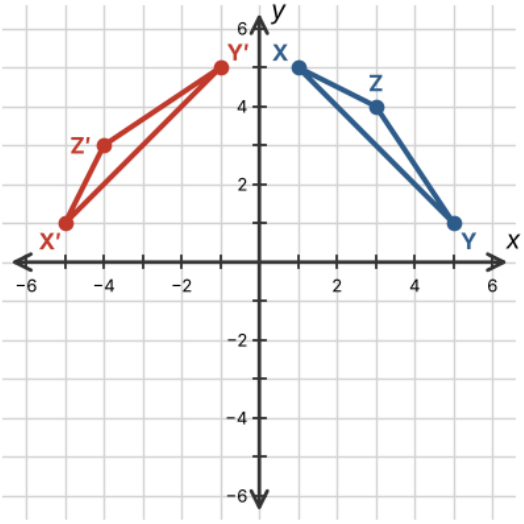
Honors Geometry – Chapter 4 Test PRACTICE WORKED KEY

10. Rotate the point (1, 5) about the origin by (3)

(a) 90° (b) 180° (c) 270°

→ **(a) (−5, 1) (b) (−1, −5) (c) (5, −1)**

11. Graph △XYZ with X(1, 5), Y(5, 1), Z(3, 4) and its image after a 90° rotation about the origin. (3)



(x, y) → (−y, x)

→ **X′(−5, 1), Y′(−1, 5), Z′(−4, 3)**

12. Glide reflection: translate by ⟨0, 1⟩, then reflect in the y-axis. Find the final image of △JKL with J(5, 5), K(4, 1), L(4, 5). (3)

middle: J′(5, 6), K′(4, 2), L′(4, 6)

→ **J″(−5, 6), K″(−4, 2), L″(−4, 6)**

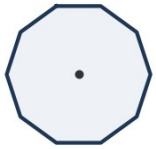
13. Name each transformation. (2)

(a) (x, y) → (−x, y)

(b) (x, y) → (y, −x)

→ **(a) reflection in the y-axis; (b) rotation of 270° about the origin**

14. The figure is a regular decagon. What is the smallest angle of rotation that maps it onto itself? How many lines of symmetry does it have? (2)



360° ÷ 10

→ **36°; 10 lines**

15. Error analysis. Dante rotated (−4, 1) 90° counterclockwise about the origin and got (1, 4). Name the mistake, then give the correct image. (2)

→ **Used the 270° rule (y, −x). 90° is (−y, x): (−1, −4)**

16. △ABC is reflected in the y-axis to form △A′B′C′. AB = 10 cm and m∠B = 42°. Find A′B′ and m∠B′. Explain in one sentence. (2)

→ **A′B′ = 10 cm, m∠B′ = 42°: a rigid motion keeps every length and angle.**

17. What is the image of (−6, −2) after a 270° rotation about the origin? (1)

(F) (2, −6) (G) (−2, 6) (H) (−2, −6) (J) (6, 2)

→ **G (−2, 6)**